Hardware & Software Requirements

- **ESP32 Development Board** (such as the DOIT ESP32 DEVKIT V1) connected to your PC via a data USB cable.
- **Arduino IDE** installed with the ESP32 board package configuration enabled.

Step 1: Configuring the Arduino Environment

1. Open the Arduino IDE, navigate to **Tools > Board > ESP32 Arduino**, and select **DOIT ESP32 DEVKIT V1**.
2. Connect your board and select its corresponding COM port under **Tools > Port** (e.g., `COM10`).
3. Set your code serial monitor communication speed to `115200` baud.

Step 2: Writing and Executing an HTTP GET Request

To download data from a web server, use the native `<WiFi.h>` and `<HTTPClient.h>` libraries. This script connects to your local 2.4 GHz network and queries a public JSON endpoint via HTTP.

[Check the HTTP_GET.ino file for code]

- **Flashing the Code:** Click the **Upload** arrow icon.
- **Viewing Output:** Open the **Serial Monitor**, verify the baud dropdown is set to **115200**, and tap the physical **EN/RST** button on the ESP32 to restart the script and view the tincoming JSON string.

Step 3: Transmitting Data via HTTP POST

To send custom packets of data (such as sensor metrics or device telemetry) to a server, transition from a `GET` request to a `POST` request by embedding a JSON payload inside the body.

[Check the HTTP_POST.ino file for the code]

- **Verifying Transmission:** An **HTTP Response Code: 201** in your Serial Monitor indicates that your packet was successfully received, parsed, and registered as a new record by the server.